

CustomerIQ

Official Product Requirements Documentation

Days 5–8 Consolidated PRD

Predictive Modeling → Model Comparison → ML Validation → Customer Risk Scoring

Purpose: Preserve the official project record for the predictive-intelligence stage of CustomerIQ.

End-of-Day-8 status: validated ML workflow completed; Gradient Boosting is the leading candidate; 3,705 eligible customers have customer-level churn probabilities and risk segments.

1. Shared Project Contract

Days 5–8 inherit the earlier customer-level dataset and churn methodology. The observed outcome is provisional: a customer is treated as churned when no purchase is observed during the defined 90-day validation window.

Item	Value
Source	UCI Online Retail II
Eligible customers	3,705 customers with 2+ pre-cutoff orders
Observation period	Transactions before 2011-09-10
Validation period	2011-09-10 through 2011-12-09 (90 days)
Predictors	Frequency; Monetary; Recency; Customer Lifespan; Average Purchase Interval; Median Purchase Interval
Excluded from predictors	Customer ID; returned_in_validation; observed_churn
Train/test split	Stratified 80/20; random_state=42

Core leakage rule: predictive features must be constructed only from information available before the cutoff; future validation behavior is used only to define/evaluate the outcome.

2. DAY 5 — Predictive Modeling

Objective

Establish an interpretable baseline model for predicting observed churn.

Method

- Logistic Regression baseline.
- StandardScaler contained inside the modeling pipeline.
- Six behavioral predictors.
- Stratified 80/20 split with random_state=42.
- Customer ID and outcome/future-information fields excluded from X.

Verified Results

Metric	Result
Train / Test	2,964 / 741
Accuracy	0.7045
Precision	0.7068
Recall	0.6272
F1	0.6646
ROC-AUC	0.7731
Strongest positive coefficient	Average Purchase Interval (0.8286)
Strongest negative coefficient	Median Purchase Interval (-0.4052)

Deliverable

- Data/Processed/logistic_regression_predictions.csv

Interpretation

The baseline demonstrated useful predictive signal and created an interpretable benchmark. It was not treated as production-ready.

3. DAY 6 — Model Comparison

Objective

Compare alternative classification approaches under the same feature and split contract.

Models

- Logistic Regression — interpretable baseline.
- Random Forest — nonlinear ensemble.
- Gradient Boosting — nonlinear boosting model.

Evaluation & Findings

Finding	Verified result
Leading candidate	Gradient Boosting
ROC-AUC	0.7769
F1	0.6990
ROC-AUC improvement vs Day 5	+0.0038
Random Forest	Large train/test gap; overfitting concern

Deliverable

- Data/Processed/model_comparison_results.csv

Interpretation

Gradient Boosting narrowly led on the tracked metrics. The improvement is limited, so this is a candidate selection for further validation, not proof of production superiority.

4. DAY 7 — ML Validation

Objective

Validate the candidate models more rigorously using stratified cross-validation and held-out test evaluation while checking leakage and model-quality issues.

Validation Design

- 3-fold Stratified Cross-Validation; random_state=42.
- Stratified 80/20 train/test split.
- Logistic Regression, Random Forest, Gradient Boosting.
- No hyperparameter tuning and no nested cross-validation.
- Random Forest: 100 estimators; Gradient Boosting: 100 estimators.

Verified Execution

Item	Result
Execution	PASS
Models	Logistic Regression; Random Forest; Gradient Boosting
Best model	Gradient Boosting
ROC-AUC	0.7769
F1	0.6990
Errors	None
Execution time	~60 seconds
Source data modified	NO
Days 1–6 modified	NO

Diagnostic outputs

- 06_confusion_matrices.png
- 07_roc_curves.png
- 08_precision_recall_curves.png
- 09_probability_distribution.png
- 10_threshold_analysis.png
- 11_feature_importance.png

Interpretation

The workflow executes successfully and Gradient Boosting remains the leading candidate. The results do not establish certainty, causality, or production readiness.

5. DAY 8 — Customer Risk Scoring

Objective

Transform the validated ML approach into a customer-level risk layer that can later feed retention workflows, Power BI, and an AI recommendation layer.

Workflow

Customer data → behavioral features → Gradient Boosting → churn probability → risk segment

Prototype Risk Rules & Results

Segment	Probability rule	Customers	Share
HIGH	≥ 0.70	690	18.62%
MEDIUM	0.40 to <0.70	1,511	40.78%
LOW	<0.40	1,504	40.59%
TOTAL	—	3,705	100%

Verified outputs

- Mean predicted churn probability: 0.465952 (~46.60%).
- 3,705 customers scored; no duplicate IDs; no missing probabilities.
- Probabilities constrained to 0–1; segments constrained to HIGH/MEDIUM/LOW.

Important business caveat

The 0.70 and 0.40 thresholds are business/prototype thresholds, not statistically optimized thresholds. Future work should test them against retention economics, campaign capacity, customer value, and observed outcomes.

Deliverables

- Notebook/08_customer_risk_scoring.ipynb
- Data/Processed/customer_risk_scores.csv

6. Data, Model & Governance Controls

Protected files

- Data/Raw/online_retail_II.csv
- Data/Processed/online_retail_cleaned.csv
- Days 1–6 notebooks and existing PRD documentation.

Current predictive artifacts

Artifact	Status
customer_churn_analysis.csv	Customer-level analysis dataset
logistic_regression_predictions.csv	Day 5
model_comparison_results.csv	Day 6
ml_validation_results.csv	Day 7
customer_risk_scores.csv	Day 8

Current limitations

- Observed churn is provisional and tied to a 90-day look-forward period.
- ROC-AUC 0.7769 indicates useful signal, not certainty.
- Risk thresholds are prototype/business thresholds.
- No causal inference is claimed.
- No production deployment, monitoring, or live customer integration has been completed.

7. Product Transition Roadmap

Layer	Purpose	Status / role
ML	Predict churn probability	Established; continue validation
Risk scoring	Prioritize customers	Day 8 complete
SQL	Operational business querying	Upcoming
Power BI	Decision dashboard and KPIs	Upcoming
AI	Explain risk and recommend actions	Later stage
Prototype / SaaS	Usable product interface	Later stage

Core CustomerIQ loop: Detect → Prioritize → Explain → Act → Measure

The product should ultimately answer: **Who is likely to churn? Why? Who should we prioritize? What should we do next?**

8. Official Completion Record — Days 5–8

Day	Status	Primary deliverable
Day 5	COMPLETE	Logistic Regression baseline + predictions
Day 6	COMPLETE	Three-model comparison + results CSV
Day 7	COMPLETE	3-fold ML validation + diagnostics + validation CSV
Day 8	COMPLETE	Customer churn probabilities + risk segmentation

Official conclusion: Days 5–8 completed the predictive intelligence foundation of CustomerIQ. The project has progressed from customer-level behavioral analysis to validated churn prediction and actionable customer risk scoring. The next priority is operationalization through SQL and Power BI, followed by AI recommendations and the product prototype.

This PRD consolidates the verified Days 5–8 results provided in the working session. It does not invent unreported metrics.